LAB EXPERIMENT RECORD

SET 17: Employee Performance

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
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SET 17: Employee Performance Analysis

Aim:

To visualize employee performance, years of service, and departmental distribution using R programs and an interactive Tableau dashboard**.**

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	**Sales**	**5**	**85**
2	**HR**	**3**	**92**
3	**Marketing**	**7**	**78**

PROGRAM 1: Line Chart (Performance over Service Tenure)

Aim: To display employee performance scores in relation to their years of service using a line chart.

R Program:

```
graphics.off()

years_of_service <- c(5, 3, 7)
scores <- c(85, 92, 78)

order_index <- order(years_of_service)

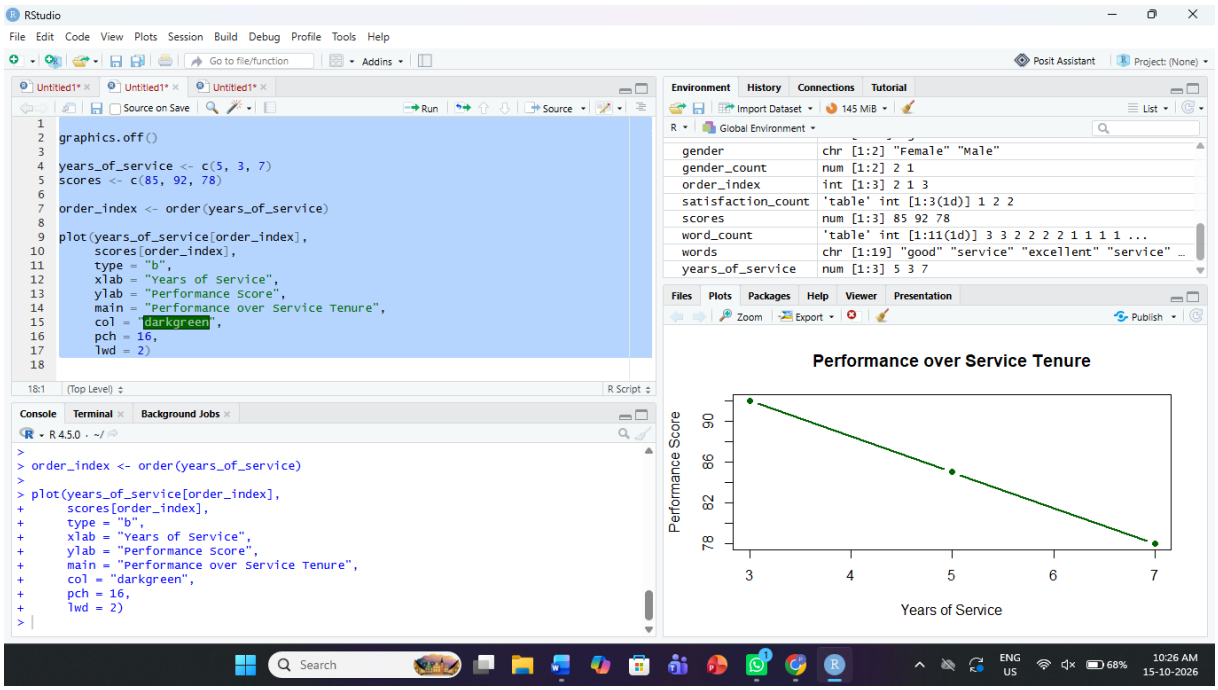
plot(years_of_service[order_index],
    scores[order_index],
    type = "b",
    xlab = "Years of Service",
    ylab = "Performance Score",
```

main = "Performance over Service Tenure",

col = "darkgreen",

pch = 16,

lwd = 2)



Observation: The chart shows performance scores against years of service. The score decreases from 92 to 85 and then to 78 as the displayed service years increase. This pattern is based on only three employees**.**

PROGRAM 2: Bar Chart (Departmental Placement)

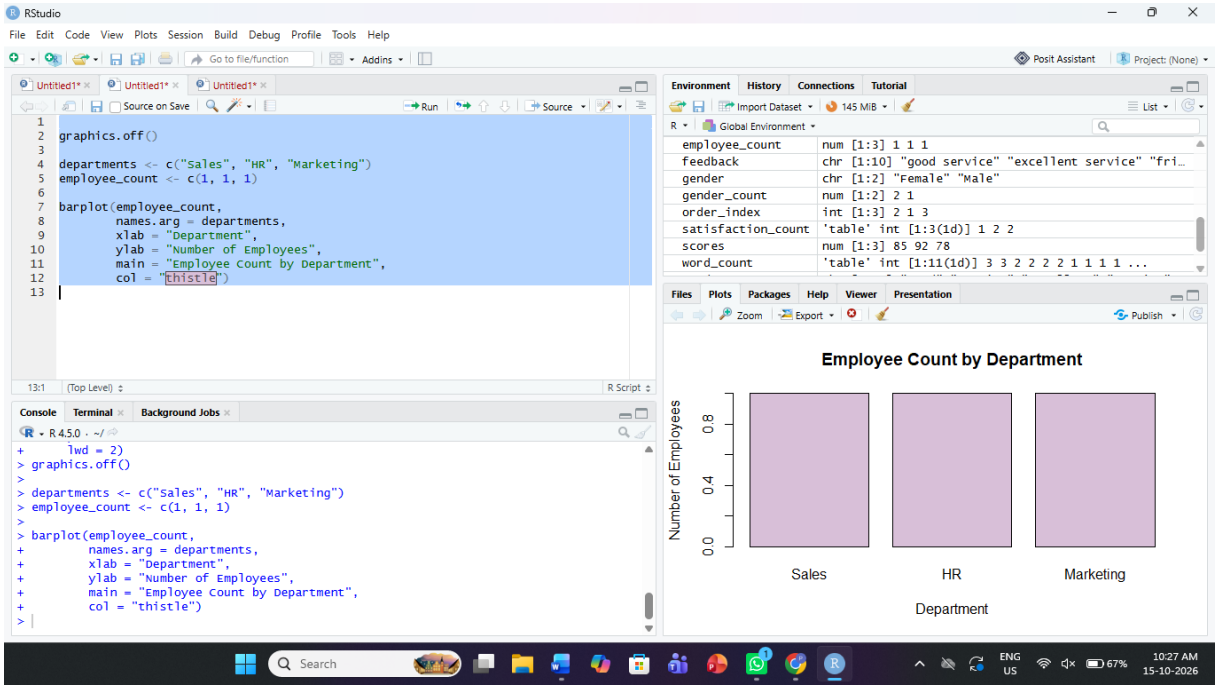
Aim: To display the number of employees in each department using a bar chart.

R Program:

```
graphics.off()

departments <- c("Sales", "HR", "Marketing")
employee_count <- c(1, 1, 1)

barplot(employee_count,
        names.arg = departments,
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Employee Count by Department",
        col = "thistle")
```



Observation: The chart shows one employee in each department. Sales, HR, and Marketing have equal employee counts in the given dataset**.**

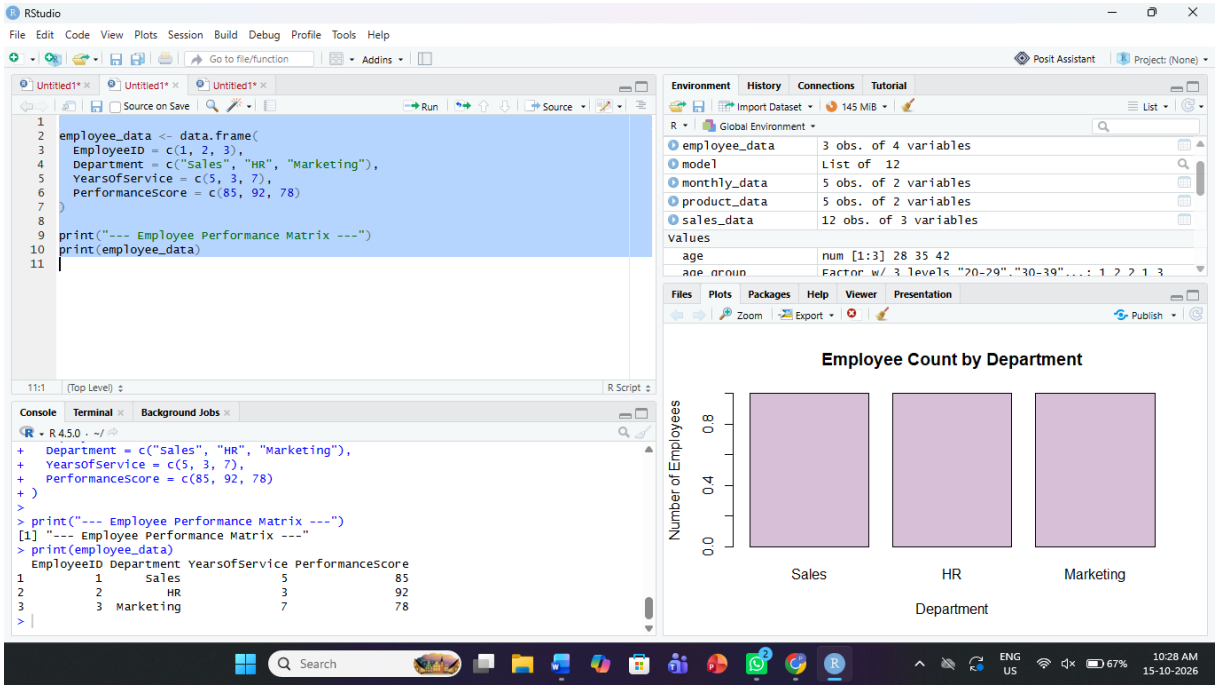
PROGRAM 3: Employee Performance Summary Table

Aim: To create a structured table summarizing employee performance records.

R Program:

```
employee_data <- data.frame(
  EmployeeID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  YearsOfService = c(5, 3, 7),
  PerformanceScore = c(85, 92, 78)
)

print("--- Employee Performance Matrix ---")
print(employee_data)
```



Observation: The table displays employee ID, department, years of service, and performance score. The HR employee has the highest performance score of 92.

PROGRAM 4: Employee Performance Dashboard in Tableau

Aim: To create an interactive dashboard combining employee performance, departmental distribution, and employee records**.**

Steps in Tableau:

1. Save the dataset as Employee_Performance.xlsx and connect it to Tableau.
2. Sheet 1 – Performance Line Chart: Drag Years of Service to Columns and Performance Score to Rows. Select Line.
3. Sheet 2 – Departmental Breakdown: Drag Department to Columns and Employee ID to Rows. Change Employee ID aggregation to Count.
4. Sheet 3 – Performance Data Table: Display Employee ID, Department, Years of Service, and Performance Score.
5. Create a new dashboard and arrange the three sheets.
6. Enable Use as Filter to filter the employee performance information.

```
graphics.off()

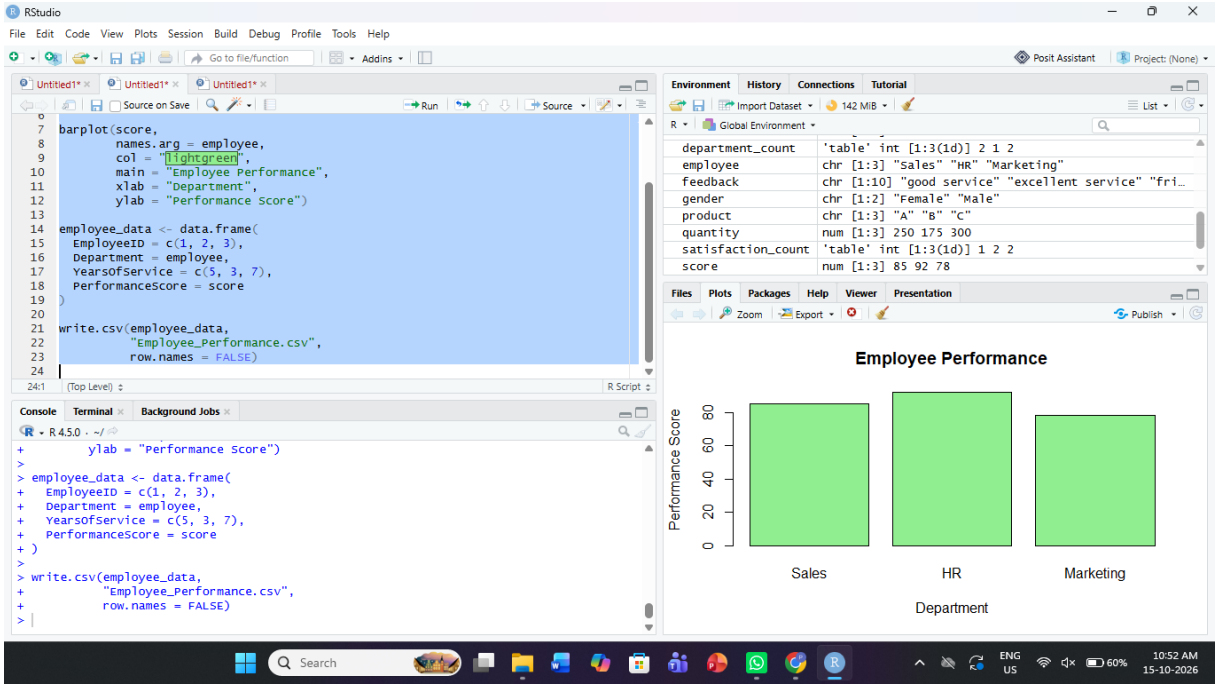
employee <- c("Sales", "HR", "Marketing")
score <- c(85, 92, 78)

barplot(score,
```

```
        names.arg = employee,
        col = "lightgreen",
        main = "Employee Performance",
        xlab = "Department",
        ylab = "Performance Score")

employee_data <- data.frame(
  EmployeeID = c(1, 2, 3),
  Department = employee,
  YearsOfService = c(5, 3, 7),
  PerformanceScore = score
)

write.csv(employee_data,
          "Employee_Performance.csv",
          row.names = FALSE)
```



Observation: The dashboard combines performance trends, department counts, and employee records. Interactive filtering helps users examine selected departments.

Procedure:

1. Create vectors containing employee details.
2. Sort employees by years of service and use plot() with type = "b" to display performance trends.
3. Use barplot() to compare department-wise employee counts.

4. Create a data frame to display employee performance records.
5. Combine the visualizations in a Tableau dashboard.

Result:

Employee performance data was visualized using a line chart, bar chart, and summary table in R.

An interactive employee performance dashboard was designed in Tableau.